Performance Comparison of CPU Scheduling Algorithms: FCFS, SJF and Round Robin

Md. Mainul Islam (ID: 23203037)
Jannati Ferdous (ID: 23303157)

A Thesis in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)



Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT - International University of Business Agriculture and Technology

Fall 2026

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The thesis has been examined and approved,

_____

Prof Dr. Utpal Kanti Das
Chairman & Professor
Dept. of Computer Science and Engineering
IUBAT - International University of Business Agriculture and Technology

_____

Dr. Hasibur Rashid Chayon
Coordinator and Associate Professor
Dept. of Computer Science and Engineering
IUBAT - International University of Business Agriculture and Technology

_____

[Supervisor Name]
[Supervisor Designation]
Dept. of Computer Science and Engineering
IUBAT - International University of Business Agriculture and Technology

Fall 2026

LETTER OF TRANSMITTAL

8th November, 2026

The Chairman
Thesis Defense Committee
College of Engineering and Technology (CEAT)
IUBAT - International University of Business Agriculture and Technology
4 Embankment Drive Road, Sector 10, Uttara Model Town
Dhaka 1230, Bangladesh

Subject: Submission of the thesis report.

Sir,

With due respect, we are pleased to submit our report titled **"Performance Comparison of CPU Scheduling Algorithms: FCFS, SJF and Round Robin"** as a part of the requirements for the Bachelor of Computer Science and Engineering (BCSE) program.

Working on this topic helped us understand how a research report is planned and written. We learned how to define a problem, set clear objectives and explain our approach in simple steps. We hope the report meets your expectations.

We would be very grateful if you kindly go through the report and give us your valuable feedback.

Thank you.

Md. Mainul Islam
ID: 23203037

Jannati Ferdous
ID: 23303157

STUDENT'S DECLARATION

We, Md. Mainul Islam (ID: 23203037) and Jannati Ferdous (ID: 23303157), are students of the Bachelor of Computer Science and Engineering (BCSE) program under the College of Engineering and Technology (CEAT) of IUBAT - International University of Business Agriculture and Technology. We declare that this report, titled **"Performance Comparison of CPU Scheduling Algorithms: FCFS, SJF and Round Robin"**, has been prepared by us as a part of our thesis course.

The work in this report is our own. Where we have used ideas or information from other sources, we have mentioned them in the references.

This report has not been submitted anywhere else for any other degree, award or presentation.

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Md. Mainul Islam	**Jannati Ferdous**
ID: 23203037	ID: 23303157

ACKNOWLEDGMENTS

First of all, we thank Almighty Allah for giving us the strength and patience to complete this report.

We would like to express our sincere gratitude to our course teacher and thesis supervisor for the guidance, advice and encouragement we received throughout this work. The clear instructions on how to plan and write a thesis report made our task much easier.

We are also thankful to the Chairman and the Coordinator of the Department of Computer Science and Engineering, IUBAT, for their support. We thank all our teachers, who taught us the basics that we used in this work.

Finally, we thank our families and friends for their love, prayers and support. Without them, this work would not have been possible.

SUPERVISOR'S CERTIFICATION

This is to certify that the thesis report titled **"Performance Comparison of CPU Scheduling Algorithms: FCFS, SJF and Round Robin"** has been prepared by Md. Mainul Islam (ID: 23203037) and Jannati Ferdous (ID: 23303157) of IUBAT - International University of Business Agriculture and Technology, as a part of the requirements of the Bachelor of Computer Science and Engineering (BCSE) program.

The report was prepared under my supervision. To the best of my knowledge and according to the declaration of the students, no part of this report has been submitted anywhere for any degree, diploma or certificate.

The students are allowed to submit the report. I wish them every success in their future work.

Thesis Supervisor

_____

[Supervisor Name]
[Supervisor Designation]
Department of Computer Science and Engineering
IUBAT - International University of Business Agriculture and Technology

DEPARTMENT'S CERTIFICATION

On behalf of the Department of Computer Science and Engineering, IUBAT - International University of Business Agriculture and Technology, we confirm that the thesis report **"Performance Comparison of CPU Scheduling Algorithms: FCFS, SJF and Round Robin"** for the Bachelor of Computer Science and Engineering (BCSE) degree was duly presented by Md. Mainul Islam (ID: 23203037) and Jannati Ferdous (ID: 23303157) and approved by the department.

_____

[Supervisor Name]
Supervisor, [Designation]
Department of Computer Science and Engineering
IUBAT - International University of Business Agriculture and Technology

_____

Dr. Hasibur Rashid Chayon
Coordinator and Associate Professor
Department of Computer Science and Engineering
IUBAT - International University of Business Agriculture and Technology

_____

Prof. Dr. Utpal Kanti Das
Chairman & Professor
Department of Computer Science and Engineering
IUBAT - International University of Business Agriculture and Technology

ABSTRACT

A computer often has many programs ready to run at the same time, but the processor can run only one of them at a moment. The part of the operating system that decides which program runs next is called the CPU scheduler. A good choice makes programs finish sooner and makes the computer feel faster, while a poor choice makes users wait for a long time.

Many scheduling algorithms exist, and each one has its own strengths and weaknesses. Students and new developers often find it hard to know which algorithm is better for a given situation. This thesis compares three well-known algorithms: First Come First Served (FCFS), Shortest Job First (SJF) and Round Robin (RR). The work is limited to a single processor and to processes that use only the CPU.

Our approach is a simple simulation. We will write a small program that takes a set of processes with their arrival times and burst times, runs each algorithm on the same set, and calculates the average waiting time, the average turnaround time and the average response time. Several sets of processes of different sizes will be used so that the comparison is fair.

This report presents the introduction chapter, which explains the background, the research gap, the research questions, the objectives, the hypotheses and the plan of the full thesis. The results will be reported in the later chapters after the simulation is completed. We expect the study to show clearly which algorithm suits which kind of workload, and to serve as an easy guide for learners.

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Chapter I. Introduction

1.1 Background of the Study

When we use a computer, many programs seem to run together. We can listen to music, browse the web and download a file at the same time. In reality, a processor with one core can run only one program at any moment. The operating system gives the processor to one program for a short time, then to another, and keeps doing this so fast that it looks like everything is running together.

A running program is called a process. The processes that are ready to run wait in a line called the ready queue. The part of the operating system that picks the next process from this queue is called the CPU scheduler. The rule that the scheduler follows to pick the next process is called a scheduling algorithm.

The choice of the algorithm matters because it changes how long each process waits and how soon it finishes. A good algorithm keeps the CPU busy and keeps the waiting time low. A poor one can make a small and urgent task wait behind a big one. For this reason, CPU scheduling is one of the first and most important topics in the study of operating systems.

1.2 Review of Related Literature

CPU scheduling is a classic topic and is explained in standard textbooks on operating systems, such as Operating System Concepts by Silberschatz, Galvin and Gagne. These books describe the common algorithms, which are First Come First Served, Shortest Job First, Priority Scheduling and Round Robin, and they explain how to judge them by waiting time, turnaround time and response time.

Many students and researchers have also compared these algorithms with small simulations. Most of these comparisons use a few sample processes and show that no single algorithm is the best in every case. For example, SJF gives the smallest average waiting time, but it can leave long processes waiting. Round Robin is fair to all processes, but its result

depends on the size of the time slice. A detailed review of selected studies will be given in Chapter II.

1.3 Overview of the Current Situation

Today, real operating systems use advanced schedulers that mix several ideas, and these are hard for a beginner to understand. The simple algorithms such as FCFS, SJF and Round Robin are therefore taught first, because they are the base of all the advanced ones. Table 1.1 gives a short summary of these three algorithms.

Table 1.1 Summary of the three scheduling algorithms

Algorithm	**Main idea**	**Known weakness**
First Come First Served (FCFS)	The process that arrives first runs first, until it finishes.	A long process at the front makes all others wait (convoy effect).
Shortest Job First (SJF)	The process with the smallest burst time runs first.	Long processes may wait too long, and burst time must be known.
Round Robin (RR)	Each process gets a small fixed time slice in turn.	Too many switches if the time slice is very small.

Table 1.1 gives the main idea and the known weakness of each algorithm.

As the table shows, every algorithm has a good side and a bad side. FCFS is the simplest and is fair in the order of arrival, but it can be slow for short processes. SJF is very good for the average waiting time, but it needs to know the burst time in advance. Round Robin gives every process a turn, but it adds extra switching work.

1.4 Research Gap and Problem Statement

Textbooks explain each algorithm well, but they usually show only one small example for each. A learner who wants to know how the three algorithms behave on different kinds of workloads, such as many short processes or a few long ones, does not find a simple side by side comparison in one place.

The problem of this study can be stated in one line: it is not easy for a beginner to see which of the basic scheduling algorithms performs better under which workload. This study tries to solve this problem with a clear and simple simulation.

1.5 Importance of the Research

The topic is important for students, because scheduling is a core part of the operating systems course and the idea is used in many other places, such as handling requests on a web server. For teachers, the study can provide ready examples with numbers that are easy to explain in a class. For beginners in software development, it shows why the choice of a rule can change the speed of a system.

The study is also simple and low in cost. It needs only a computer and a small program, so anybody can repeat it and check the results.

1.6 Research Questions

This thesis tries to answer the following questions:

1. How do FCFS, SJF and Round Robin differ in average waiting time and average turnaround time?
2. How does the average response time of Round Robin compare with that of FCFS?
3. How does the size of the workload change the performance of each algorithm?

1.7 Aim and Objectives

The main aim of this thesis is to compare the performance of the FCFS, SJF and Round Robin scheduling algorithms by simulation. To reach this aim, we set the following objectives:

- To study how each of the three algorithms works.
- To write a simple simulation program that runs each algorithm on the same set of processes.
- To calculate the average waiting time, turnaround time and response time of each algorithm.

- To repeat the test with different numbers of processes and different time slices.
- To find out which algorithm suits which kind of workload.

1.8 Hypotheses

A hypothesis is a statement that can be tested with data. Based on the research questions and the theory in the textbooks, we make the following hypotheses for this study:

- H1: SJF gives a lower average waiting time than FCFS and Round Robin.
- H2: Round Robin gives a lower average response time than FCFS.

We will test both hypotheses with the simulation results. If the results do not support a hypothesis, we will report it honestly and try to explain the reason.

1.9 Scope and Definition of Terms

The scope of this study is limited to three algorithms: FCFS, SJF (non-preemptive) and Round Robin. We assume a single processor and processes that need only the CPU, with no waiting for input or output. We do not study priority scheduling, multi-core systems or real operating system schedulers. These can be taken up in future work.

Some terms used in this report are explained below to make the reading easy.

- Process: a program that is running or is ready to run.
- Arrival time: the time at which a process enters the ready queue.
- Burst time: the total time a process needs on the CPU.
- Waiting time: the total time a process spends in the ready queue.
- Turnaround time: the time from the arrival of a process to its completion.
- Response time: the time from the arrival of a process to its first turn on the CPU.



Figure 1.1 Basic idea of CPU scheduling

Figure 1.1 shows that processes wait in the ready queue, the scheduler picks one of them by the chosen rule, and the CPU runs it until it finishes or its turn ends.

1.10 Outline of the Methodology

This study will use a simulation method. First, we will study the three algorithms and write down the steps of each one. Second, we will write a small program in a common language such as Python or C that follows these steps. Third, we will prepare several sets of sample processes, each with an arrival time and a burst time, in different sizes such as 5, 10 and 20 processes.

Finally, we will run each algorithm on every set, calculate the three average times, and show the results in tables and charts. For Round Robin, we will try more than one time slice. The full details of the method will be given in Chapter III.

1.11 Organization of the Thesis

This thesis is planned in five chapters. Chapter I is the introduction, which explains the background, the research gap, the questions, the objectives and the hypotheses. Chapter II is the literature review, which discusses earlier studies and shows the gap that our work tries to fill. Chapter III explains the research methodology in detail. Chapter IV presents the results and discusses them. Chapter V concludes the thesis and gives some ideas for future work. The references are given at the end of the report.